

EXPERIMENT - 16

BINARY SEMAPHORE IN SYSTEMC

CODE:

```
#include <systemc.h>

SC_MODULE(binary_semaphore) {
    int sem;

    sc_event ev;

    void task1() {
        while(true) {
            wait(1, SC_SEC);
            if(sem == 1) {
                sem = 0;
                cout << "Task1 Entered Critical Section at " << sc_time_stamp() <<
endl;
                wait(2, SC_SEC);
                sem = 1;
                ev.notify();
            }
            else {
                wait(ev);
            }
        }
    }

    void task2() {
        while(true) {
```

```

        wait(2, SC_SEC);

        if(sem == 1) {
            sem = 0;
            cout << "Task2 Entered Critical Section at " << sc_time_stamp() <<
endl;

            wait(1, SC_SEC);
            sem = 1;
            ev.notify();
        }
        else {
            wait(ev);
        }
    }
}

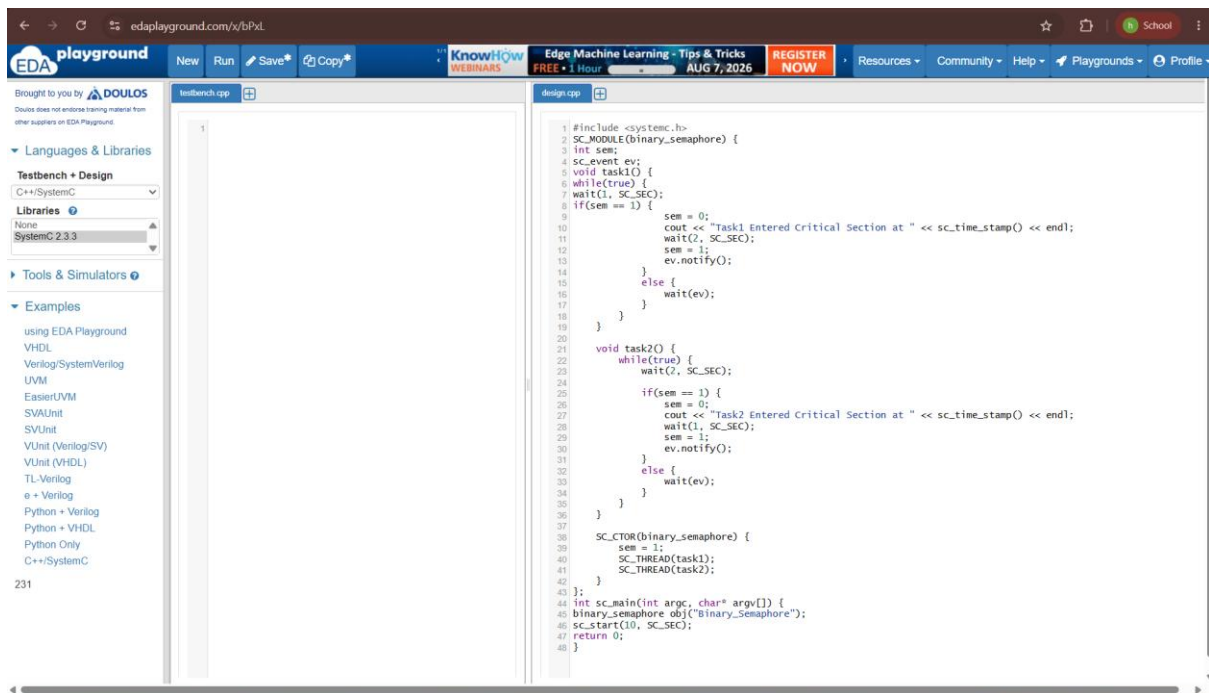
SC_CTOR(binary_semaphore) {
    sem = 1;
    SC_THREAD(task1);
    SC_THREAD(task2);
}

};

int sc_main(int argc, char* argv[]) {
    binary_semaphore obj("Binary_Semaphore");
    sc_start(10, SC_SEC);
    return 0;
}

```

SIMULATION:



OUTPUT:

